

Homework 4

Write a `HW4.Rmd` file to answer these questions. Knit your `HW4.Rmd` to `HW4.html` along the way. Turn in `HW4.html` to Canvas.

1. A high school student applies to seven colleges, each of which makes an admission decision by rolling a fair six-sided die and admitting a student if a 1 is rolled. What is the probability the student will be admitted to exactly two colleges? (Round your answer to two decimal places.)

2. To celebrate their 30th birthdays, brothers Mario and Luigi of the Nintendo *Mario* video game franchise wish to study the distribution of heights of their mushroom enemies, the Goombas. Their reasoning is that shorter Goombas are easier to jump on. (Goombas die when Mario and Luigi jump on them.)
 - (a) If we assume that the population of Goomba heights are normally distributed with mean 12 inches and standard deviation 6 inches, what is the probability that a randomly chosen goomba has a height between 13 and 15 inches?
 - (b) Koopa Troopas, other enemies of Mario & Luigi, have normally distributed heights with mean 15 inches and standard deviation 3 inches. What is the probability that a randomly chosen Koopa Troopa is taller than 75% of Goombas?

3. Let F be an RV that represents the operating temperature in Fahrenheit of one instance of a manufacturing process, and let $F \sim N(90, 5^2)$. Let C be an RV that represents the same process, but measured in Celsius. Fahrenheit can be converted to Celsius using $C = \frac{5}{9}(F - 32)$.
 - (a) Find the probability that one randomly selected instance of the process will have operating temperature greater than 93.8 Fahrenheit.
 - (b) C is also normally distributed. Use the properties of $E()$ and $VAR()$ to find its mean and variance.
 - (c) Find the probability that one randomly selected instance of the process will have operating temperature below 29 Celsius.
 - (d) Find the Celsius temperature x such that the probability that the operating temperature in Celsius of one instance is less than x is .25.